

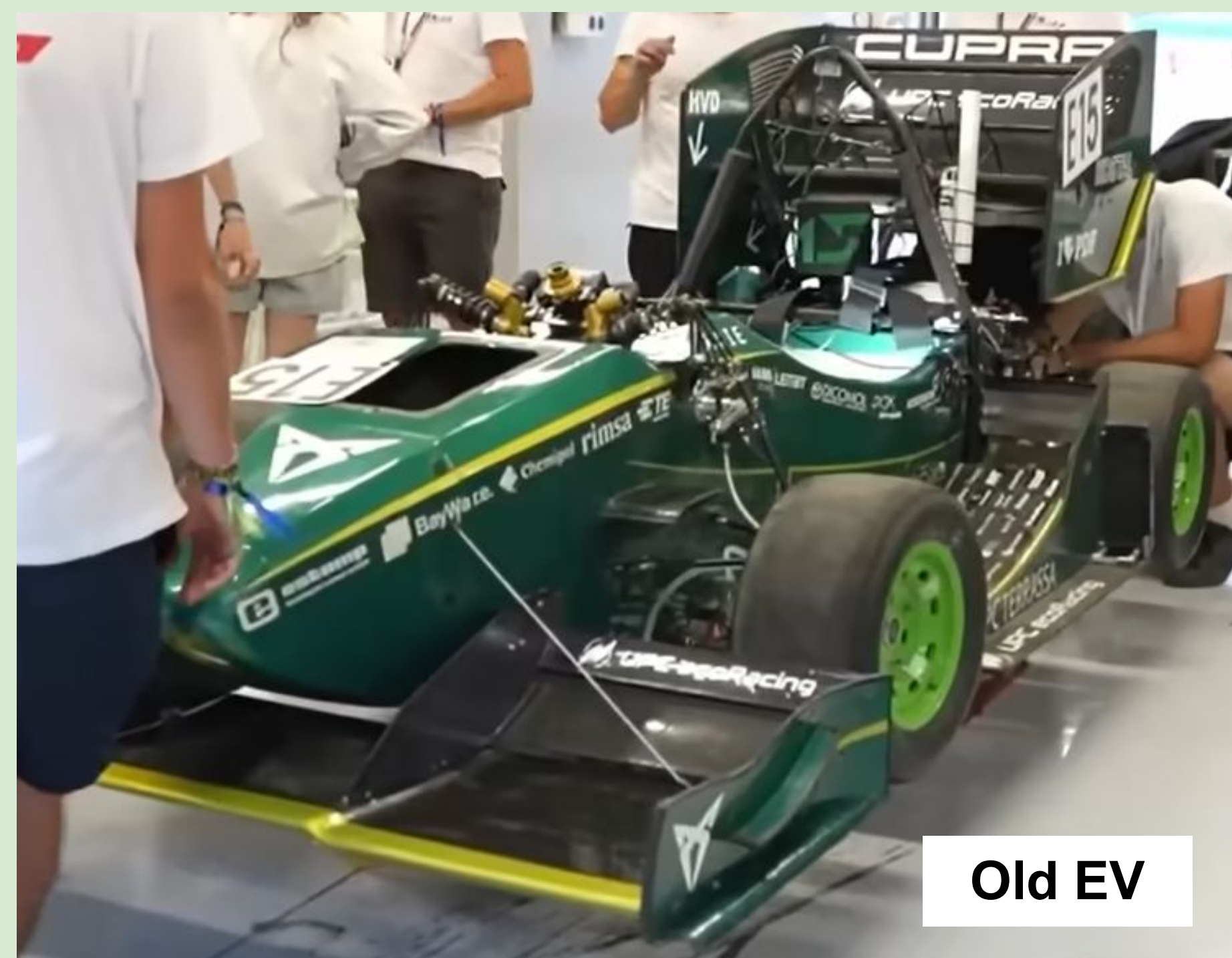


OBJECTIVE

Develop a safe, modular BMS prototype platform for Viking Motorsports' 2027 Formula SAE EV, supporting cell monitoring, fault detection, and scalable vehicle integration.

BACKGROUND

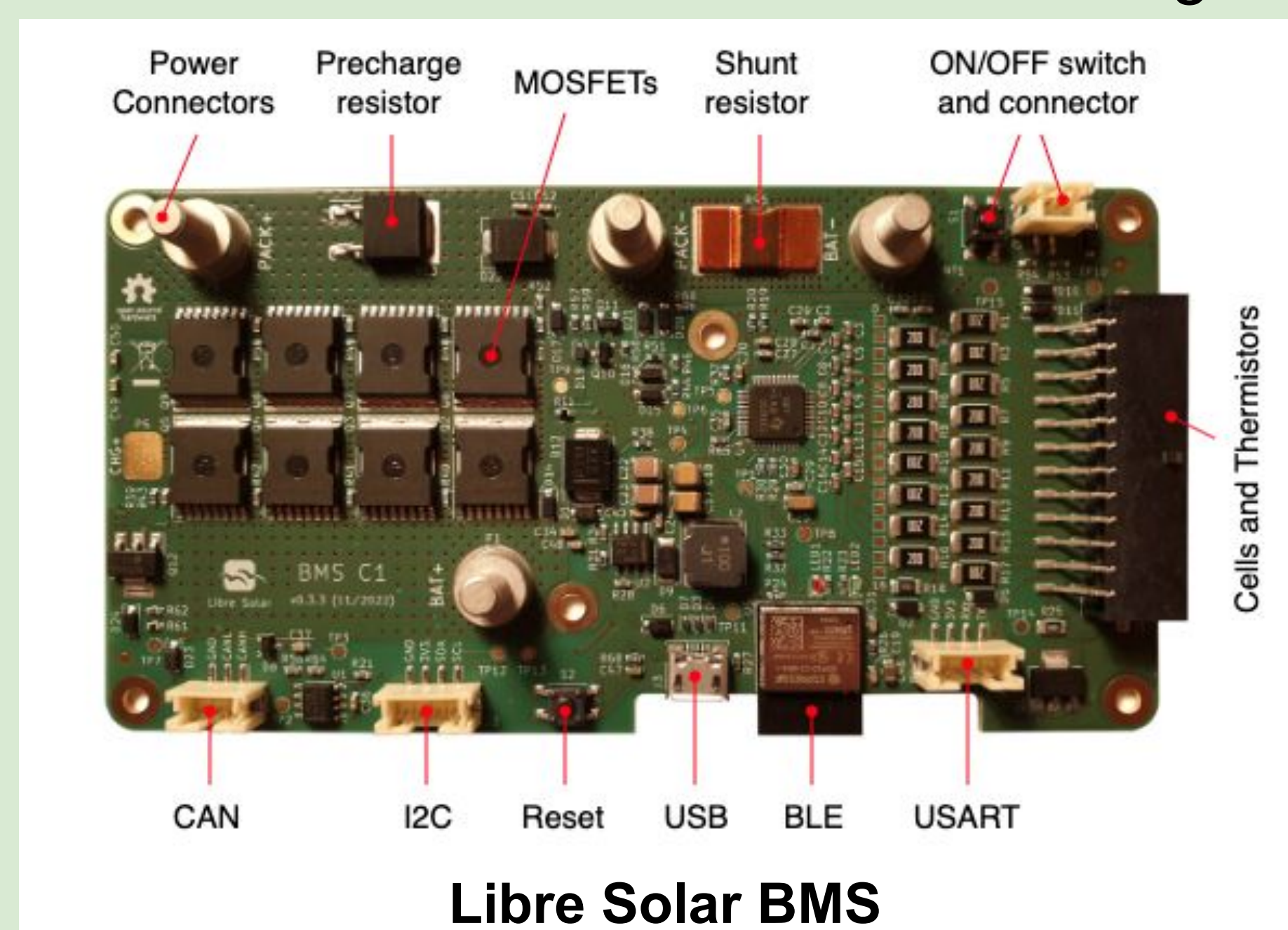
- VMS is rebuilding after retiring its previous EV
- The old centralized Li-ion NMC pack created weight distribution and modularity challenges
- The next EV requires a safer, more modular battery architecture
- No pre-existing BMS platform was available
- Formula SAE EV rules require reliable monitoring and shutdown capability



Old EV

EXISTING SOLUTIONS

- Used open-source BMS designs to guide system architecture and IC selection
- Referenced Libre Solar Cell Monitors for CMU design direction



Libre Solar BMS

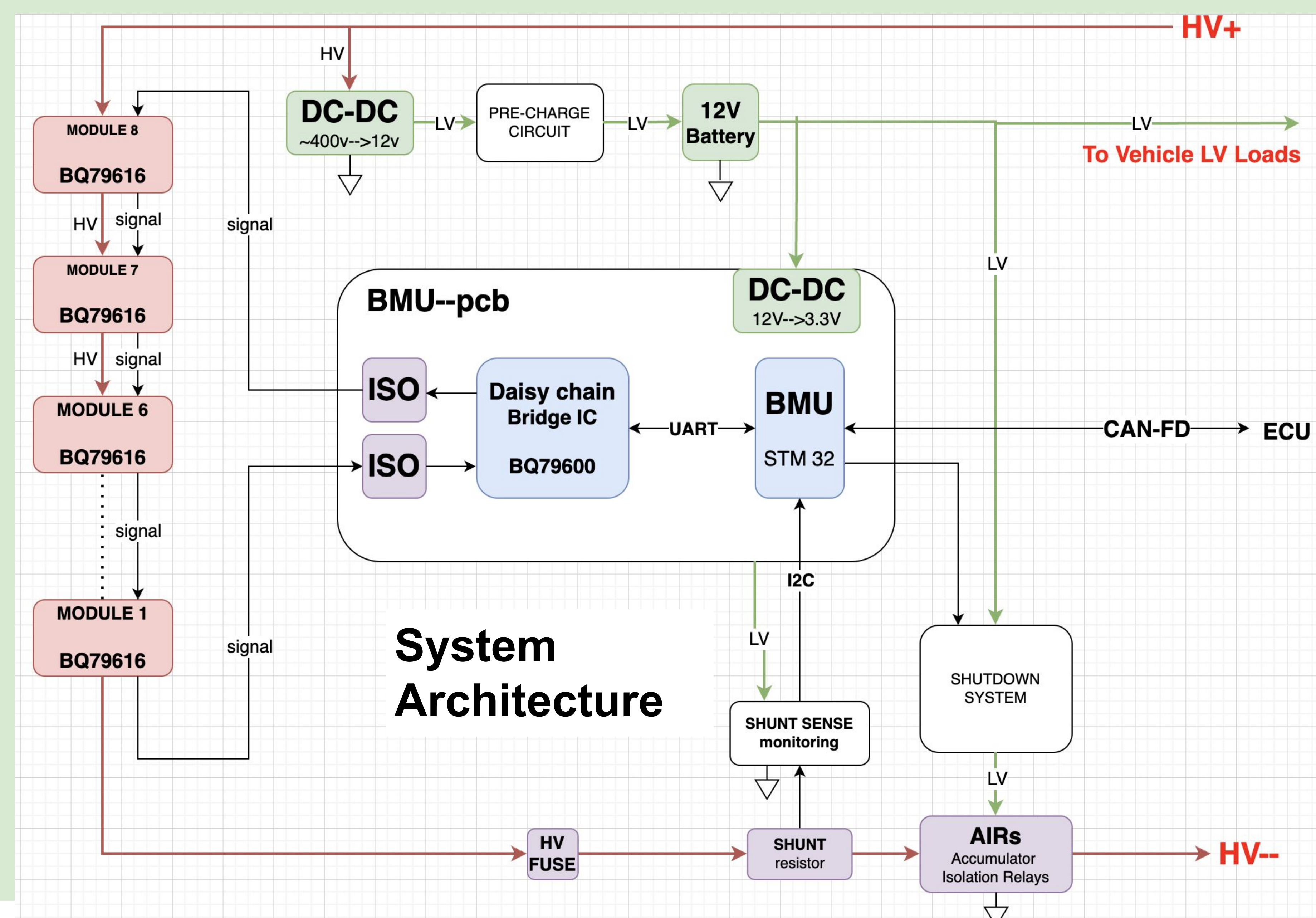
APPROACH

Modular approach in three parts:

- **Battery Management Unit (BMU)** — Coordinates the system and reports data over CAN-FD
- **Cell Monitoring Unit (CMU)** — Monitors and balances up to 16 LFP cells
Prototype target: 12 cells, $V_{max} < 48\text{ V}$
- **Pack Current Monitor** — Measures pack current using a shunt resistor

Goals:

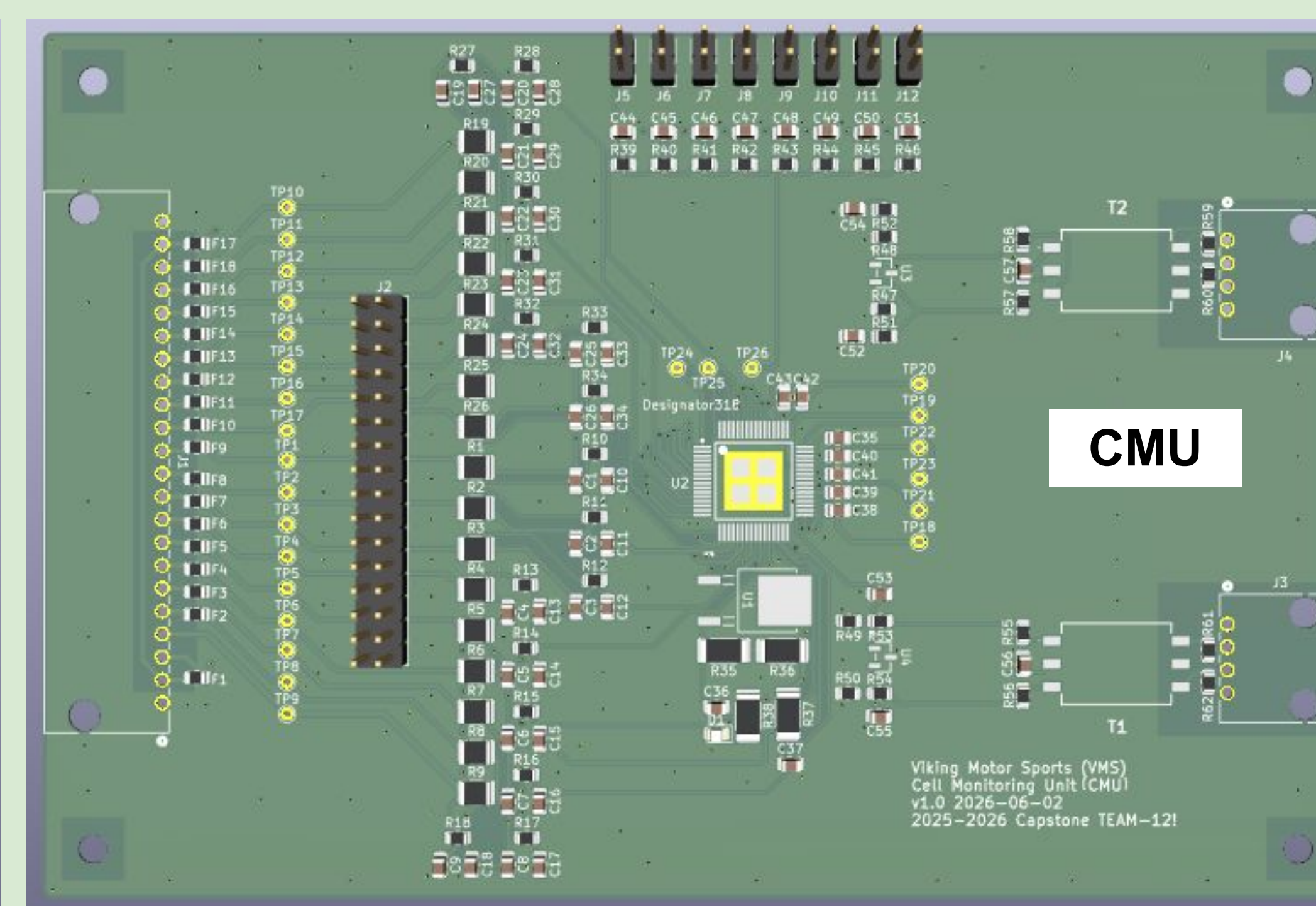
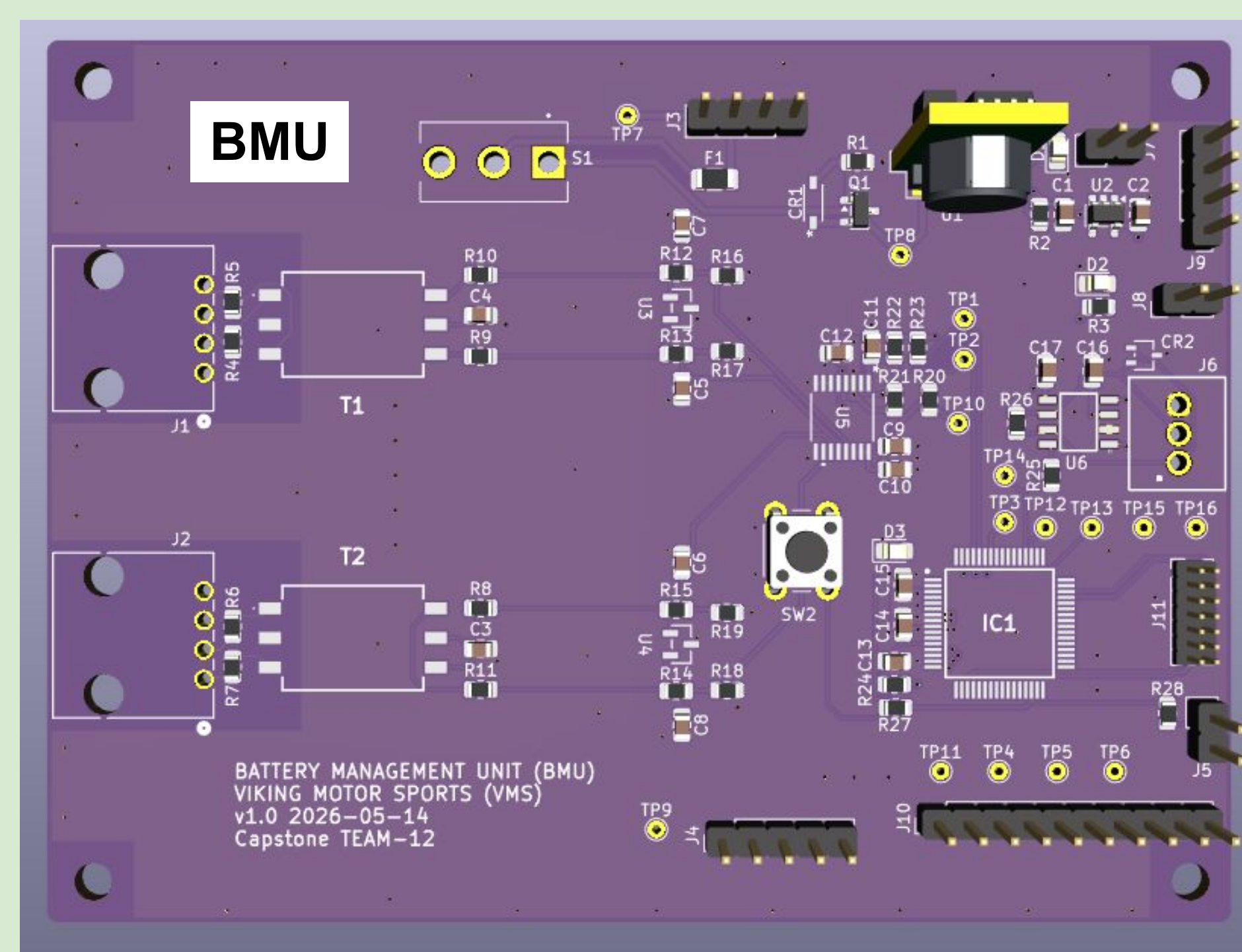
- Build a functional BMU + CMU + current monitor prototype
- Demonstrate connectivity with the Vehicle Control System (VCS)
- Document the design for future teams to build the full system
- Stretch Goal: move from evaluation boards to custom prototype PCBs



System Architecture

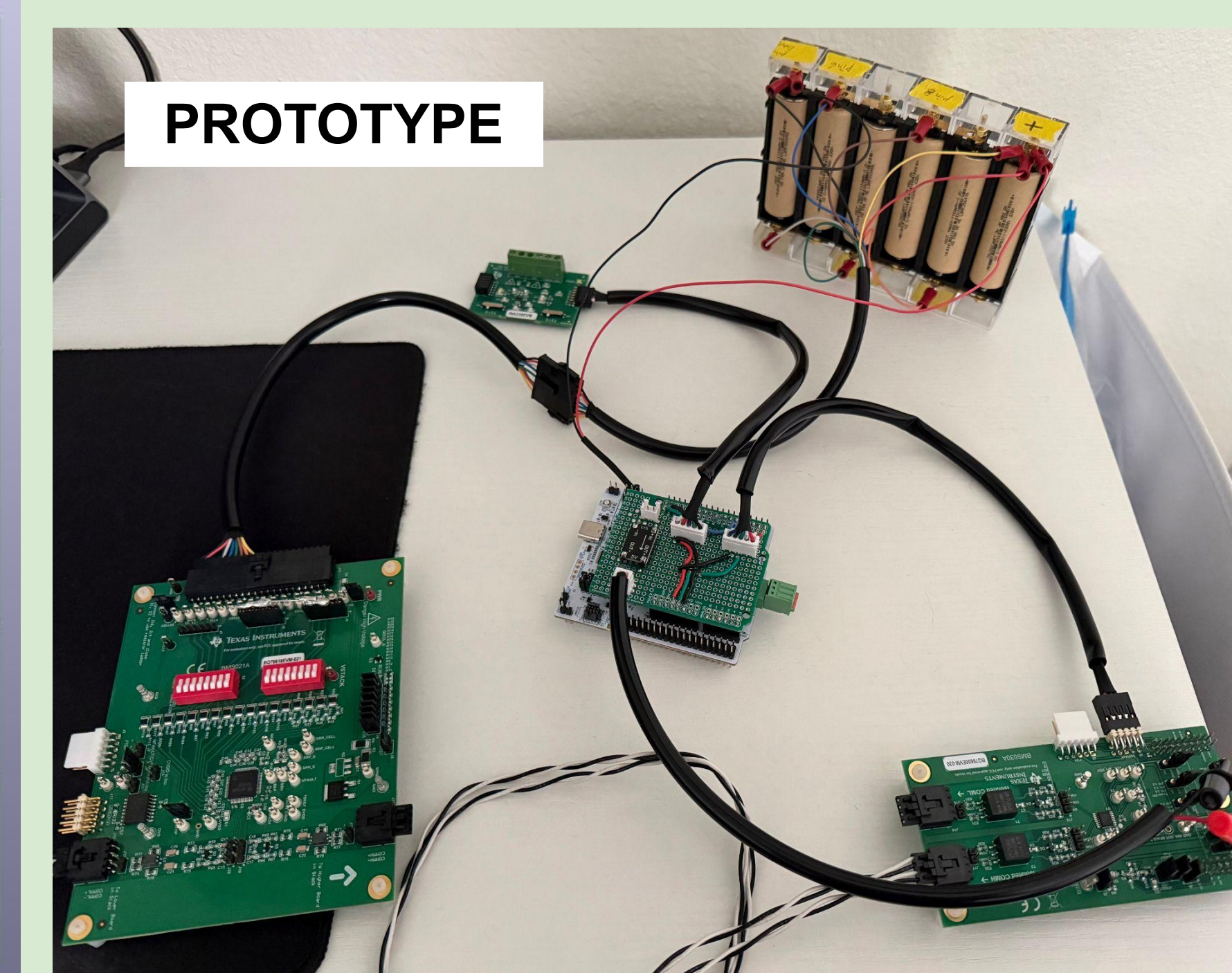
PROTOTYPING and RESULTS

- Functional evaluation-board prototype completed
 - Most firmware functions work to specification
 - One CMU connected and tested
 - Integrated with the VCS team's prototype
- Custom PCB development required more time than expected
 - BMU, CMU, and current monitor layouts are complete
 - Waiting on assembled PCBs
 - Fault detection and measurement error handling need improvement



CONCLUSION and FUTURE WORK

- First development iteration is complete
 - Only one CMU connected and tested
- Next iteration:
 - Assemble and debug custom PCBs
 - Improve measurement accuracy and fault detection
 - Expand from one CMU to a multi-CMU system
- Start development early
 - Too much time spent researching battery pack design



PROTOTYPE